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Microwave permittivity and permeability experiments in high-loss dielectrics: Caution with implicit Fabry-Pérot resonance for negative imaginary permeability

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Various materials have been examined to obtain permittivity and permeability values required for detailed investigations on functional materials like microwave-absorbing and high-permittivity dielectrics. We call for caution when samples exhibit negative imaginary permeability, whether containing a resonance peak or not. In the retrieval procedure, the Fabry-Pérot resonance (FPR) can produce a negative imaginary permeability, which should be classified as an extrinsic rather than an intrinsic physical attribute. In particular, for high-loss materials, the implicit FPR would bring a plausible negative imaginary permeability. Here we have listed a dielectric dispersion of FPR behavior for $BiFeO_3$ samples. © *2013 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4825378]

The electromagnetic properties of microwave media including metamaterials,[1–6] nanomaterials,[7–12] and magnetoelectric composites[13–15] have attracted increasing attention in recent years. Many extraordinary electromagnetic constitutive parameters were reported by the S-parameter retrieval or Nicolson-Ross-Weir (NRW)[16–18] methods based on transmission and reflection data from a finite-thickness homogenous slab. In particular, the negative imaginary permeability (μ'') is found not only in metamaterials[19] but also in natural materials/composites, such as Fe_3O_4/SnO_2 nanorods,[20] MWCNT/wax,[21] SiCw/wax,[22] and hollow cobalt nanochains composites.[23] Although the negative imaginary permittivity (ε'') and μ'' in metamaterials arouse heated debates,[24–28] it has been theoretically proven by Markel[29] that the μ'' can be negative in diamagnetics. Nevertheless, caution is still needed with the negative μ'' obtained by NRW in high-loss materials, as well as for high dielectric permittivity related to ferroelectric polarization.[30] The report by Maglione *et al.* has clearly distinguished differences between true bulk polarization and extrinsic conductivity contributions. Indeed, the NRW method has provided a convenient approach to determine the constitutive properties of materials to understand wave propagation phenomena in materials. However, the retrieved results usually suffer from Fabry-Pérot resonance (FPR) effects, which lead to counterfactual medium parameters. In particular, when a high-loss material is of interest, the permeability would be found to have negative imaginary parts due to the implicit FPR (no FPR peak is observed in the frequency range investigated). In this letter, we show that a negative μ'' can be misleading in $BiFeO_3$ (BFO) multiferroics because of the implicit FPR, which is taken in the report as an example of extrinsic behavior. Also, the temperature and geometrical response of FPR are discussed.

The BFO powders were prepared by a sol-gel method. Typically, Bi $(NO_3)_3 \cdot 5H_2O$, and Fe $(NO_3)_3 \cdot 9H_2O$ were used as starting materials and the procedure was similar to that presented in our previous report.[31] The as-prepared BFO powders were compacted into a rectangular sheet (22.82 mm $\times$ 10.12 mm $\times$ 2.56 mm), fitting the geometrical dimensions of the waveguide sample holder. The rectangular samples were sintered at 883 K for 20 min. The relative complex permittivity and magnetic permeability of the samples were determined from the S parameters by the standard NRW method in the X band (8.2–12.4 GHz). The S parameters were measured using a network analyzer (Anritsu37269D) or were obtained by finite difference time domain simulations.

Figure 1(a) shows the microwave permeability spectra. There, μ'' is negative between 9.3 and 12.4 GHz, although no explicit resonance peak is observed in the real part (μ') or imaginary part of the permeability at the frequencies investigated. With increasing frequency, μ'' decreases rapidly when the frequency is higher than 11.3 GHz, and the minimum value of μ'' is -0.14 at 12.4 GHz. Meanwhile, the corresponding ε'' increases rapidly with increasing frequency (Fig. 1(b)), and the value of ε'' is up to 3.98 at 12.4 GHz. Negative μ'' values have also been reported in other natural materials, such as Fe_3O_4/SnO_2 nanorods,[20] MWCNT/wax,[21] and hollow cobalt nanochain composites.[23] To verify whether μ'' is intrinsically negative, we measured the temperature dependence of the complex permittivity and permeability (see Fig. 2). Both the real parts of the permittivity ε' and ε'' were found to increase as the temperature increased. Moreover, μ' shows a positive temperature coefficient, especially when the frequency is higher than 10 GHz. Nevertheless, μ'' decreases with increasing temperature. This negative temperature dependence of μ'' is more obvious at higher frequencies, suggesting that the negative μ'' could not be an intrinsic property of BFO diamagnetics.

To analyze the reasons behind the negative μ'', we investigated whether thickness affects the dielectric properties of

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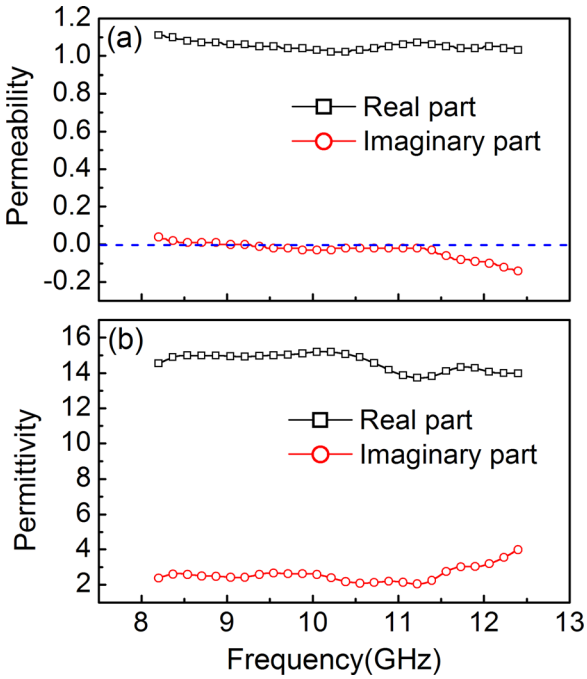


FIG. 1. The experimental microwave permeability (a) and permittivity (b) of BFO in the X band.

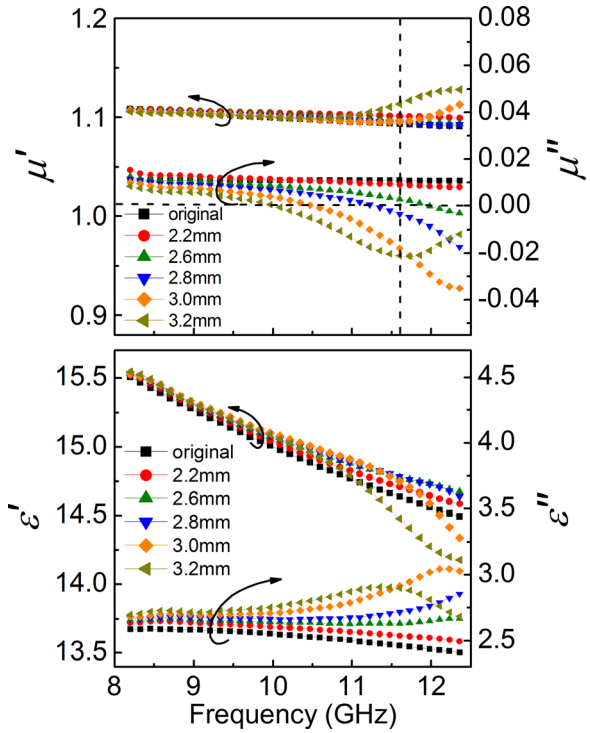


FIG. 3. The simulating microwave permeability (a) and permittivity (b) of BFO with various slab thicknesses.

samples by simulating the reflection (S_{11}) and transmission (S_{21}) spectra for homogenous slabs of various thicknesses. The variation of complex permeability and complex permittivity, calculated from the S parameters by the NRW method, was observed at different sample thicknesses (see Fig. 3). The frequency dependence of μ' and μ'' at various thicknesses (Fig. 3(a)) clearly shows that thickness causes the negative μ''. In samples of thickness 2.2 mm, all the values of μ'' are positive in the X band, and both μ' and μ'' are consistent with the original setting data. Nevertheless, in samples 2.6 mm thick, negative μ'' appears at high frequencies. The thicker the sample is, the more obvious negative behavior is present in μ'' at lower frequencies. The complex permittivity also shows similar sample-thickness-dependent character (Fig. 3(b)). The calculated values of ε' and ε'' deviated further away from the original values as sample thickness increased. Thus, we infer that size effects lead to counterfactual medium parameters. Remarkably, an explicit resonance peak was observed at 11.72 GHz (Fig. 3) (i.e., the FPR frequency), where the sample thickness is 3.2 mm which is just half of the FPR wavelength. For the single TE_{10} mode of the waveguide in the X band, the waveguide wavelength can be expressed as

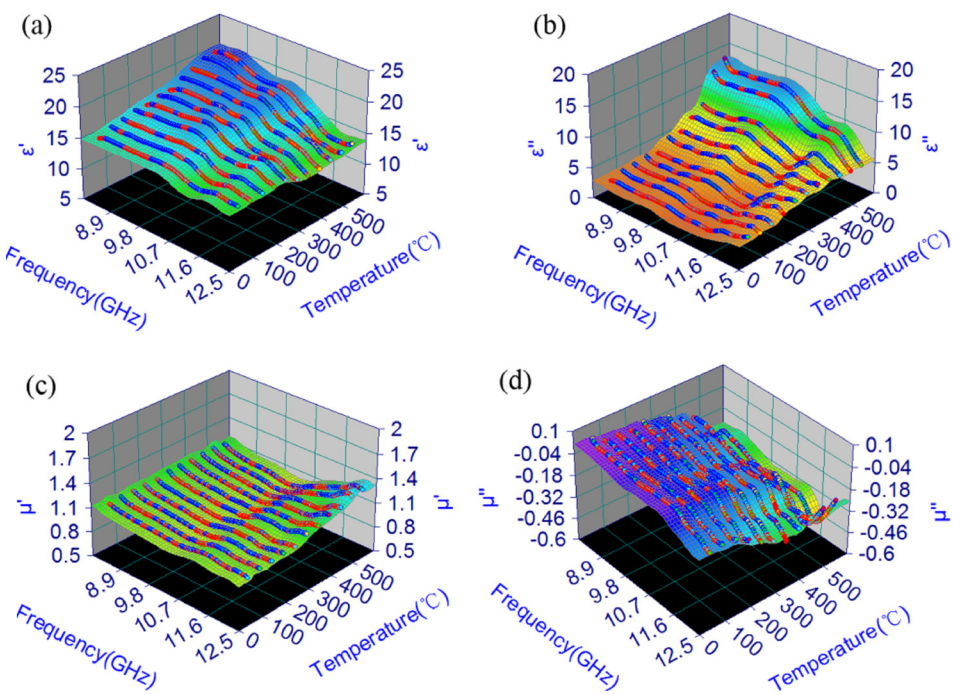


FIG. 2. The temperature and frequency dependence of real permittivity (a), imaginary permittivity (b), real permeability (c), and imaginary permeability (d) of BFO.

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$$\lambda_g = 1/\mathrm{Re}\left[\left(\frac{\varepsilon_r \mu_r f^2}{v_0^2} - \frac{1}{\lambda_c^2}\right)^{1/2}\right]. \quad (1)$$

Here, $\lambda_c = 2 \times 22.86\,\mathrm{mm} = 45.72\,\mathrm{mm}$ is the cutoff wavelength of TE_{10} mode (if the mode is TEM, $\lambda_c = \infty$), and v_0 velocity of light, f frequency, ε_r relative complex permittivity, and μ_r relative complex permeability. If the sample thickness is $\lambda_g/2$, the FPR will occur, which can manipulate the reflection of electromagnetic field.[32,33] In accordance with Eq. (1), the FPR frequency has been calculated (Table I). Obviously, the diminishing FPR frequency with increased thickness is responsible for the sample-thickness-dependent feature (Fig. 3). Note that the negative μ'' can be obtained by implicit FPR, although the FPR peak is not observed in the frequency range investigated, such as for samples of thickness 2.6, 2.8, and 3.0 mm in Fig. 3. This result confirms that the negative μ'' in Fig. 1(a) is not an intrinsic property of BFO that is caused by the implicit FPR.

Because of the FPR effect, selecting an appropriate thickness is important if correct dielectric parameters based on the NRW method are to be obtained. For magnetic materials, the optimal thickness of the samples should be $\lambda_g/4$ at the minimum frequency in the investigated range. A smaller thickness would result in much more error, whereas a larger thickness would lead to counterfactual resulting from FPR. For nonmagnetic materials, the distortion of the result caused by FPR can be adjusted by normalizing the permeability. If a rectangular slab of material with a thickness d is installed in a rectangular air-filled waveguide, the S parameters can be expressed as follows:

$$S_{11} = \frac{\Gamma(1-T^2)}{1-\Gamma^2 T^2}, \quad (2)$$

$$S_{21} = \frac{T(1-\Gamma^2)}{1-\Gamma^2 T^2}, \quad (3)$$

where T is the propagation factor $T = e^{-j\frac{2\pi}{\Lambda}d}$ with $\mathrm{Re}(\frac{1}{\Lambda}) = \frac{1}{\lambda_g}$, λ_g the transmission line guide wavelength, and Γ the reflection coefficient. Combining Eqs. (2) and (3), we have

$$\Gamma = \frac{1 + S_{11}^2 - S_{21}^2 \pm \sqrt{(S_{11}^2 - S_{21}^2 - 1)^2 - 4S_{21}^2}}{2S_{11}}, \quad (4)$$

$$T = \frac{S_{21}}{1 - \Gamma S_{11}} = e^{-j\frac{2\pi}{\Lambda}d}, \quad (5)$$

where the signs can be determined by $|\Gamma| \le 1$. According to Eq. (5), we get

TABLE I. FPR frequencies with various sample thickness.

Thickness (mm)	Resonance frequency (GHz)
2.2	17.10
2.6	14.47
2.8	13.44
3.0	12.54
3.2	11.72

$$\frac{1}{\Lambda} = \frac{1}{2\pi d}\left\{\mathrm{Im}\left(\ln\frac{1}{T}\right) + 2m\pi - i\mathrm{Re}\left(\ln\frac{1}{T}\right)\right\}, \quad (6)$$

where m is an integer related to the number of half wavelengths that propagate inside the slab. The complex permittivity and permeability can be determined from Λ and Γ

$$\mu_r = \frac{1+\Gamma}{\Lambda(1-\Gamma)\sqrt{\frac{1}{\lambda_0^2} - \frac{1}{\lambda_c^2}}}, \quad (7)$$

$$\varepsilon_r = \frac{\left(\frac{1}{\Lambda^2} + \frac{1}{\lambda_c^2}\right)\lambda_0^2}{\mu_r}, \quad (8)$$

where λ_0 is the wavelength in free space, and λ_c the cutoff wavelength of the transmission line. When the sample thickness is an integer multiple of half of the wavelength inside the sample, the results from Eqs. (6) and (7) will be heavily affected by FPR. In fact, the FPR affects Γ rather than Λ. Therefore, we can obtain the intrinsic dielectric properties of nonmagnetic materials only from Λ. If we set the nonmagnetic normalization $\mu_r = 1$, the deviation of ε' and ε'' can be compensated, and then Eq. (8) yields

$$\varepsilon_r = \left(\frac{1}{\Lambda^2} + \frac{1}{\lambda_c^2}\right)\lambda_0^2. \quad (9)$$

Figure 4 shows a comparison of the routine-calculated permittivities for nonmagnetic materials, both compensated and original. Clearly, the routine-calculated values of ε' and ε'' are found to deviate from the original in the higher frequency region because of a potentially implicit FPR at higher frequency. The obvious difference between the routine-calculated and original parameters usually results in an inaccurate understanding of the material dielectric properties. However, the nonmagnetic values after normalization of the permeability are close to the original values. Hence, the distortion of the results in nonmagnetic materials caused by FPR can be compensated by permeability normalization.

In summary, the retrieval procedure of the S parameters cannot reveal the intrinsic dielectric properties of materials when FPR is present. In particular, for high-loss materials or composites, the implicit FPR would bring a plausible

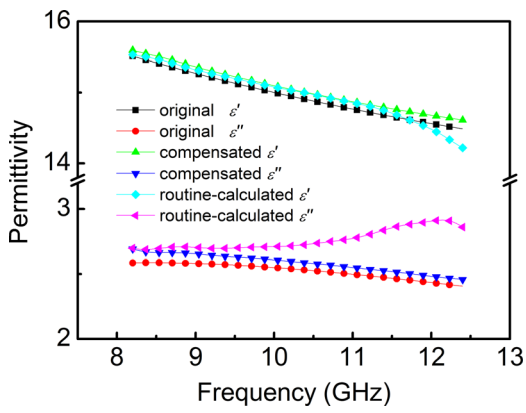


FIG. 4. Comparison of the routine-calculated, compensated, and original permittivities for nonmagnetic materials.

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negative imaginary permeability, which means that an inappropriate thickness of slab can greatly change the reflection coefficient and thus result in counterfactual dielectric parameters. We suggest that the previous reports on negative μ'' for natural materials should be reconsidered by changing the thickness of samples. The optimal slab thickness for the magnetic materials should be $\lambda_g/4$ of the minimum frequency in the investigated region. Moreover, the distortion of the result induced by FPR in nonmagnetic materials can be adjusted by permeability normalization.

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